AI-Powered Expert Advisor

Transforming Industrial Robot Support with LLM Technology

Product Manager | Industrial Robotics Division

Part 1

Product Definition

Your Customer

Field Technicians

Age:	20–60 years
Education:	85% high school, 35% some college
Languages:	English, Spanish, Cantonese, Mandarin, Urdu
Devices:	95% mobile, 75% desktop, 25% tablet
Daily work:	Configure robots, maintain uptime, troubleshoot faults

Today's Pain Points

- 7,500 articles in English only — no search filters
- Average query takes 15–30 min with no guarantee of resolution
- Complex issues require escalation to scarce senior staff
- Non-English speakers face an additional barrier
- Production downtime costs $10–50k per hour in idle lines

The Problem

What technicians need

- Instant answers to robot faults
- Step-by-step repair guidance
- Part numbers without errors
- Help in their own language
- Works on mobile in the field

How they do it today

- Keyword search across 7,500 English articles
- Ask a senior colleague (if available)
- Call company support hotline
- Read paper manuals
- Wait for escalation — hours or days

Biggest problems

- Slow: 15–30 min per query
- English-only knowledge base
- Senior staff are bottlenecks
- Escalation = production downtime
- Zero multilingual support

Business problem: Technicians cannot quickly and reliably get the guidance they need — production stops, senior staff burn out, and non-English speakers are left behind.

The Solution

RoboAdvisor: a multilingual, RAG-powered conversational assistant that gives industrial robot technicians immediate, step-by-step guidance grounded in official manuals, safety protocols, and part catalogs — accessed on any device, in any language, at any time.

Instant answers

Natural-language Q&A replaces slow keyword search. Average resolution time drops from 30 min to under 2 min.

Multilingual

Responds in English, Spanish, Cantonese, Mandarin, and Urdu — meeting every technician where they are.

Safety-first

Every response is grounded in official documents. A safety layer flags dangerous instructions before they reach the user.

Always available

On-device cache keeps core guidance available during network interruptions — no connectivity, no problem.

vs. today's patchwork of manual keyword search, unavailable experts, and multi-hour escalations

Risks & Mitigations

HIGH

Safety-critical hallucinations

Mitigation: RAG grounds every answer in verified docs. Safety review layer blocks responses lacking explicit source citation. Human escalation path always available.

HIGH

Incorrect part numbers

Mitigation: Part catalog stored as structured data (not free text). LLM retrieves part numbers from catalog via exact lookup, never generates them from memory.

MED

Adoption failure

Mitigation: Pilot with 3 sites, 2 languages. Measure weekly active users from day 1. Collect in-session thumbs-up/down feedback; iterate monthly.

MED

Network interruptions on the factory floor

Mitigation: On-device cache stores the 500 most-common procedures. Full RAG pipeline resumes when connectivity restores.

MED

Multi-language quality gaps

Mitigation: Translation verified by native-speaker technicians before launch. BLEU score and user satisfaction tracked per language.

Part 2

System Details

System Attributes

Primary Goal

Enable any field technician — regardless of language, device, or experience level — to resolve robot faults faster and more safely by providing accurate, source-grounded, conversational guidance in real time.

Secondary Goals

Zero safety incidents

Never provide instructions that could injure a technician or damage equipment. All responses cite source documents.

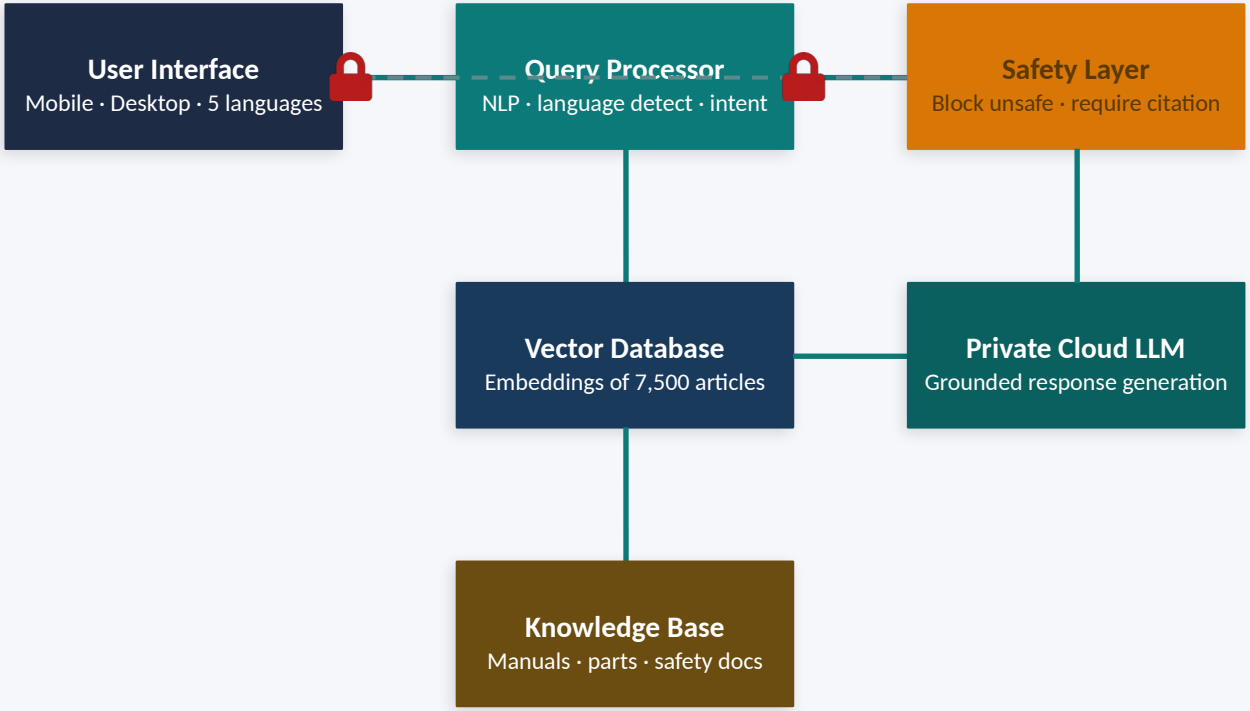
Multilingual access

Deliver accurate guidance in 5 languages to eliminate the English-only barrier faced by 30%+ of users.

Reduce escalations 60%

Resolve the majority of Tier 1 & 2 queries without contacting company support staff.

System Architecture — RAG Pipeline



🔒 All arrows within the private cloud use TLS 1.3 encryption. No technician data or query content is transmitted to third-party services.

Component notes

User Interface:
Mobile-first PWA; offline cache for top 500 procedures

Query Processor:
Detects language, classifies intent, routes to correct doc corpus

Vector Database:
Stores sentence embeddings; similarity search returns Top K=3 chunks

Private Cloud LLM:
Dedicated server; no data leaves company perimeter

Safety Layer:
Rules engine + LLM secondary check; blocks if no citation found

Knowledge Base:
Ingestion pipeline updates weekly; version-controlled

LLM Configuration

Property	Value	Rationale
License type	Proprietary (e.g., Azure OpenAI GPT-4o)	Enterprise SLA, EU AI Act compliance, vendor-managed security patches. Open-source considered but ruled out: industrial liability requires audited model provenance.
Deployment model	Dedicated private-cloud server (company VPC)	Technician queries contain proprietary fault data and machine configs. Dedicated server guarantees data never reaches shared inference infrastructure.
Temperature	0.1	Near-deterministic output. Safety-critical instructions must not vary across identical queries. Low temp prevents "creative" but wrong answers.
Top K (retrieval)	3 document chunks	Returns the 3 most relevant procedure chunks. Fewer = lower hallucination risk; more chunks risk introducing contradictory guidance.
Max output tokens	512	Sufficient for step-by-step repair instructions. Prevents verbose responses that overwhelm technicians on mobile.
Context window	16k tokens	Accommodates full retrieval context (3 × ~400-token chunks) plus conversation history for multi-turn diagnostics.
Fine-tuning	Optional — Phase 2	Initial RAG handles knowledge via retrieval. Fine-tuning considered after 6 months of user query data to improve tone and domain phrasing.
System prompt	Safety-first persona	Instructs model to always cite source, refuse speculation, escalate when uncertain, and never generate part numbers from memory.

Part 3

Measuring Success

Success Metrics

Avg. resolution time
Outcome

< 2 min

Baseline: 15–30 min

Escalation-free rate
Outcome

> 70%

Baseline: ~20% today

User satisfaction score
Outcome

> 4.2 / 5

Baseline: No baseline

Safety incidents from AI guidance
Safety

0 incidents

Baseline: Target: zero

Weekly active technicians
Adoption

> 80% of users

Baseline: 0 (new product)

Median response latency
Performance

< 3 seconds

Baseline: N/A

Sample Prompts

System Prompt

```
You are RoboAdvisor, a technical support assistant for
industrial robots.

Rules:
1. Answer ONLY from the retrieved documents provided in
context.
2. Always cite the source document name and section
number.
3. If the retrieved context does not contain a clear
answer, say:
   "I don't have a verified answer for this. Please
escalate to support."
4. NEVER generate part numbers from memory — retrieve
them from the parts catalog only.
5. Warn the user about all relevant safety precautions
before any repair step.
6. Respond in the same language as the user's question.
```

User Prompt (example)

```
My UR10e robot is showing fault code E-411. The arm
stops mid-cycle and the teach pendant displays a joint
limit warning. What should I check first?
```

Expected assistant behavior

Retrieves UR10e fault code reference + joint limit procedure. Cites "UR10e Service Manual §7.3". Lists safety checks first. Provides steps 1–4. Gives part number only from catalog lookup. Offers to escalate if steps don't resolve fault.

Reference Data

Knowledge base content

- 7,500 service articles (English)
- 45,000 part records with descriptions
- 15 robot model manuals (PDF)
- Safety & regulatory docs
- Translated to 4 languages (Phase 1)

Data quality & bias checks

- Native-speaker review per language
- BLEU score ≥ 0.75 for translations
- SME sign-off on safety procedures
- Version control — no deprecated docs in index
- Quarterly accuracy audit vs. support tickets

Ingestion & refresh

- Weekly pipeline: ingest → chunk → embed → index
- Change detection flags updated articles
- New robot models added within 30 days of launch
- Technician feedback loop flags missing content
- Soft delete for deprecated parts — never hard delete

The Opportunity

Every minute of downtime costs $10–50k. RoboAdvisor can cut resolution time by 90%.

90%	60%	5
Reduction in resolution time	Fewer escalations to senior staff	Languages served from day one

Questions?